

UmiPlus DataCollection SDK User Guide

Connect to the Raspberry Pi of UmiPlus via SSH

Bash

```
1 ssh user@ip
```

Clone the SDK, enter the collection directory, create the environment and activate it.

Bash

```
1 git clone https://github.com/giaoWH/UmiPlus_DataCollection.git
2 cd ./UmiPlus_DataCollection
3 conda create -n umi-sdk python=3.11 -y
4 pip install -r requirements.txt
```

Find Ports:

1. Check data output directory: UmiPlus_DataCollection/config/record.yaml->output_root
2. Enter commands in the terminal: `python scripts/sdk_discover_ports.py --scope serial`

Please follow the instructions in the program strictly during operation.

```
● (base) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ conda activate umi_sdk
○ (umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_discover_ports.py --scope serial
=== 串口自动发现 ===
将按以下顺序识别串口：
  1. FT -> ft.port
  2. IMU -> imu.port
  3. Motors -> motors.port
以下已启用设备不参与串口发现：realsense, microphone, camera, gelsight
识别过程中，已识别成功的设备可以保持连接。
请先拔掉所有待识别的串口传感器，然后按 Enter 开始...|
```

- a. Unplug these three USB interfaces first, then press Enter.

- b. Plug in FT and press Enter.
 - c. Plug in IMU and press Enter.
 - d. Plug in motor control board and press Enter.
 - e. The process is completed. The program will automatically save data and exit.
3. Enter commands in the terminal: `python scripts/sdk_discover_ports.py --scope visual`

The program will automatically recognize RealSense without manual intervention.

Select the one with 640x480@30Hz when choosing the camera.

Select the 3280x2464@25Hz option when choosing the Gelsight.

```
(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_discover_ports.py --scope visual
[ WARN:0@4.852] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video9): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
ioctl(VIDIOC_QBUF): Bad file descriptor
[ WARN:0@5.281] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video10): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.282] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video11): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.284] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video12): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.285] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video13): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.286] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video14): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.287] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video15): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device

准备为 Camera 选择 device_index
当前候选视频设备：
- index=6 name=U3344W: U3344W frame=640x480 fps=30.0
- index=8 name=GelSight Mini R0B 2BU9-WE83: Ge frame=3280x2464 fps=25.0
- index=9 name=GelSight Mini R0B 2BU9-WE83: Ge frame=640x480 fps=90.0
请输入 device_index; 输入 'p <index>' 预览, 'r' 重新扫描, 's' 跳过, 'q' 退出: 
```

```
(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_discover_ports.py --scope visual
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.286] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video14): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device
[ WARN:0@5.287] global cap_v4l.cpp:914 open VIDEOIO(V4L2:/dev/video15): can't open camera by index
[video4linux2,v4l2 @ 0xaaaae51374e0] Not a video capture device.
[video4linux2,v4l2 @ 0xaaaae51374e0] ioctl(VIDIOC_G_INPUT): Inappropriate ioctl for device

准备为 Camera 选择 device_index
当前候选视频设备:
- index=6 name=U3344W: U3344W frame=640x480 fps=30.0
- index=8 name=GelSight Mini R0B 2BU9-WE83: Ge frame=3280x2464 fps=25.0
- index=9 name=GelSight Mini R0B 2BU9-WE83: Ge frame=640x480 fps=90.0
请输入 device_index; 输入 'p <index>' 预览, 'r' 重新扫描, 's' 跳过, 'q' 退出: 6

准备为 GelSight 选择 device_index
当前候选视频设备:
- index=8 name=GelSight Mini R0B 2BU9-WE83: Ge frame=3280x2464 fps=25.0
- index=9 name=GelSight Mini R0B 2BU9-WE83: Ge frame=640x480 fps=90.0
请输入 device_index; 输入 'p <index>' 预览, 'r' 重新扫描, 's' 跳过, 'q' 退出: 8

视觉设备发现完成, 已写回配置文件。
- realsense: auto
- camera: 6
- gelsight: 8
已生成备份文件: /home/wanghao/Desktop/UMI_DataCollection_media_exp/configs/record.yaml.bak
```

Start collection:

1. Enter commands in the terminal `python scripts/sdk_record.py`, the program will start calibration automatically. **Please ensure the device is completely stable, stationary and placed upright on the desktop during this process.** A prompt will appear once calibration is finished.

```
(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_record.py
=== SDK 交互式录制启动中 ===
配置文件: /home/wanghao/Desktop/UMI_DataCollection_media_exp/configs/record.yaml
数据源: real
interactive=true 时将忽略 duration_sec=10.0
interactive=true 时将忽略 duration_sec=10.0
[FT_Sensor] 进程已启动
[IMU] 进程已启动
[realsense] 线程已启动
[Motors] 进程已启动
[microphone] 线程已启动
[camera] 线程已启动
[gelsight] 线程已启动
等待传感器就绪...
[FT_Sensor] 零点校准完成, 基线: F=[-7.609, 4.802, 3.568] T=[-0.3721, -0.0176, -0.0431]
[Motors] 零点校准完成, 基线: [1.282, -0.0081, 0.2099, 0.376, -0.0087, 0.08]
丢弃启动阶段数据: 1.0 秒

所有传感器已就绪, 校准完成。
操作说明: 每填写一次 task name, 将连续录制并保存 5 条; 首条仍然是第一次空格填写 task name, 第二次空格开始录制, 第三次空格结束录制
录制结束后按 Enter 保存当前 session, 按 q 放弃当前 session, 按 Ctrl+C 退出
```

2. Press the space bar to enter the task name. Chinese characters are supported. Press Enter to confirm.

Note: The task name entered this time will take effect for the next n pieces of data, and n pieces of data will be collected per task. You will be prompted to enter a new task name after collection is finished. The value of n can be configured via the parameter **task_name_batch_size** in `/config/record.yaml`.

```

所有传感器已就绪，校准完成。
操作说明：每填写一次 task name，将连续录制并保存 5 条；首条仍然是第一次空格填写 task name，第二次空格开始录制，第三次空格结束录制
录制结束后按 Enter 保存当前 session，按 q 放弃当前 session，按 Ctrl+C 退出

=== Session Annotation ===
请输入 task name（支持中文），按 Enter 确认。空值或 Esc 取消本次准备。
task name> 拿起红色的3d打印零件
已记录 task name: 拿起红色的3d打印零件
按空格开始录制，按 q 取消本次准备，按 Ctrl+C 退出。

```

3. Press spacebar to start collection, press it again to stop. Press Enter to save after stopping, press q to discard if this data sample fails.
4. After saving or discarding, press the spacebar again to start or stop, and repeat the process.
5. After finishing collection, press Ctrl + C and the program will exit automatically.

When should we discard this segment of data?

1. Operational errors during collection:
 - a. The gripper accidentally touches irrelevant objects multiple times.
 - b. Failure to hold firmly causes severe unintended shaking of the device.
 - c. Other issues affecting image clarity.
2. Device errors occurred:
 - a. All-zero rows appear in the data summary table after disk writing.

Possible Problems:

1. Encounter [Realsense] runtime error: Frame didn't arrive within 1000

```

(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_record.py
=== SDK 交互式录制启动中 ===
配置文件: /home/wanghao/Desktop/UMI_DataCollection_media_exp/configs/record.yaml
数据源: real
interactive=true 时将忽略 duration_sec=10.0
interactive=true 时将忽略 duration_sec=10.0
[FT_Sensor] 进程已启动
[IMU] 进程已启动
[realsense] 线程已启动
[Motors] 进程已启动
[microphone] 线程已启动
[camera] 线程已启动
[gelsight] 线程已启动
等待传感器就绪...
[realsense] 运行时错误: Frame didn't arrive within 1000
[realsense] 运行时错误: Frame didn't arrive within 1000
[FT_Sensor] 零点校准完成, 基线: F=[-7.55, 4.829, 3.561] T=[-0.3733, -0.014, -0.043]
[Motors] 零点校准完成, 基线: [1.288, -0.0075, 0.185, 0.376, -0.0088, 0.1005]
丢弃启动阶段数据: 1.0 秒
[realsense] 运行时错误: Frame didn't arrive within 1000

```

Reconnect the wiring of the RealSense camera.

2. Problem as shown in the figure below:

```
(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ python scripts/sdk_record.py
=== SDK 交互式录制启动中 ===
配置文件: /home/wanghao/Desktop/UMI_DataCollection_media_exp/configs/record.yaml
数据源: real
interactive=true 时将忽略 duration_sec=10.0
interactive=true 时将忽略 duration_sec=10.0
[FT_Sensor] 进程已启动
[IMU] 进程已启动
[realsense] 线程已启动
[Motors] 进程已启动
[microphone] 线程已启动
Segmentation fault (core dumped)
(umi_sdk) wanghao@wh-pi:~/Desktop/UMI_DataCollection_media_exp$ [FT_Sensor] 零点校准完成, 基线: F=[-6.6
66, 5.247, 3.478] T=[-0.3789, -0.0083, -0.0372]
[Motors] 零点校准完成, 基线: [1.284, -0.0091, 0.2, 0.3755, -0.0077, 0.09]
```

Restart the device by entering `sudo reboot` in the terminal. After the device restarts, reconnect it with your computer and go through the discover port process again.

3. If stuck waiting for sensors to be ready, press Ctrl+C to exit first. If it fails to exit, press Ctrl+\. Then redo the process of discover port --serial.
4. Failed to connect to VS Code server: Connect to Raspberry Pi via SSH in terminal, then delete `~/ .vscode`

Bash

```
1 rm -rf ~/ .vscode
```

Then reconnect using VS Code. It will automatically redownload the VS Code server, and you can use it normally once the download is complete.

5. If you hear any abnormal noise, gear slipping, or feel any unusual resistance, first check every screw on the transmission chain.
6. 【Sensor Runtime Error】: Press Ctrl+C to stop the collection program, open a new terminal, enter `conda activate umi_sdk` to activate the environment, then rerun `python scripts/sdk_record.py`.
7. In case of Gelsight errors, prioritize powering off for cooling. Resume power supply and data collection after 2 to 3 minutes. After powering on again, you need to rediscover ports and reconnect the RealSense camera.

Unified Solutions for Unexpected Issues:

1. Press and hold the power button of the Raspberry Pi until the indicator light next to the power button turns red.
2. Force power off.
3. Close all remote connection terminals on your laptop.
4. Power off the device (wait for one minute).
5. Power on the device.
6. Wait for the device to reconnect to your computer's hotspot.
7. Connect to the device via terminal or VS Code.
8. Navigate to the project directory and activate the Python environment.
9. Discover ports, detect cameras, and reconnect the RealSense camera.
10. Perform static calibration.

Troubleshooting Solutions:

1. Reconnect all cables
2. Rediscover ports then re-plug the RealSense camera
3. Restart the collection program
4. Reboot the Raspberry Pi
5. Power off and restart after cooling down